$$S(n,j)=G(n)-\sum_{i=1}^{j-1}f(p[i])+\sum_{i=j}^{p[i]^2<=n}\sum_{e=1}^{p[i]^{e+1}<=n}(f(p[i]^e)S(\lfloor\frac{n}{p[i]^e}\rfloor,i+1)+f(p[i]^{e+1}))$$